

HEAT CHANGES IN REACTION (THERMOCHEMISTRY)

INTRODUCTION

It is noticed that energy in the form heat (thermal energy) is generally evolved or absorbed as a result of a chemical change. Thermochemistry is the branch of physical chemistry which deals with the thermal or heat changes caused by chemical reactions.

We have studied in the previous topic that every substance has a definite amount of energy known as the intrinsic energy or internal energy, E . Its exact value cannot be determined but the change in internal energy, ΔE , can be accurately measured experimentally.

When the internal energy of the reactants (E_r) is > internal energy of the products (E_p), the difference is internal energy, ΔE , is released as heat energy.

$$\Delta E = E_{\text{products}} - E_{\text{reactants}}$$

or $\Delta E = E_p - E_r$

Such reaction is called exothermic reaction. If the $E_p > E_r$, heat is absorbed from the surroundings. Such is called endothermic reaction. The amount of heat released or absorbed in a chemical reaction is termed the heat of reaction.

The energy changes in chemical reactions are largely due to the breaking of existing bonds between the atoms and the formation of new bonds.

CAUTION OF A REACTION

The change in internal energy (ΔE) is the heat change accompanying a chemical reaction at constant volume because no external work is performed. However, at constant pressure not only does the change in internal

energy takes place but work is also involved because of expansion or contraction.

The enthalpy of a system is defined as the sum of the internal energy and the product of its pressure and volume, that is,

$$H = E + pV$$

The value of enthalpy and internal energy will depend on the temperature and pressure for any given physical state. In order to standardise the data in heats of reaction, it is necessary that some reference point be established for the thermodynamic state of the reactants and products. The choice of such a reference point is just arbitrary and is called standard state of the substance.

In thermochemistry, temperature is 25°C (298K) and pressure is 1 atm and generally used as standard conditions for all substances (pure solids, liquids and ideal gases). More recently, the standard is defined at pressure of 1 atm at very specified temperatures. The symbols g , l , and s are used in the parentheses after the formula to denote whether the substance taking part in the reaction is gas, liquid or crystalline solid, respectively.

Heat of Reaction

It is the change in enthalpy when molar quantities of reactants shown by the balanced equation for the reaction react completely.

for a constant positive process.

[illegible]

for reactions involving only liquids and solids, rate of $\bar{v}$ very slight. PAV is usually negligible in comparison

with ΔE_0 . for reactions involving gases,

$$P \Delta V = \Delta n RT.$$

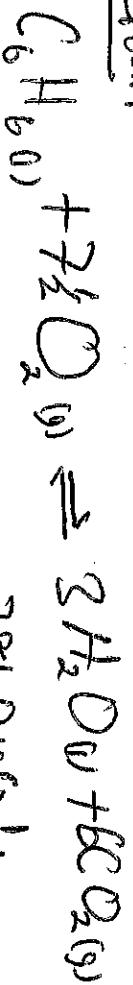
$$\therefore (\Delta H)_p = \Delta E + \Delta n RT \quad \text{---} \quad (2)$$

where Δn is the change in the number of moles of gaseous products and reactants (assumed, gases are ideal).

Example:

When 1 mole of liquid benzene is completely burnt in oxygen to form liquid water and carbon dioxide gas, ΔH is -781.0 kcal at 25°C . Calculate the heat of this reaction at constant volume at the same temperature.

Solution:



$$\Delta H_{298} = -781.0 \text{ kcal}.$$

The number of moles n_r of gaseous reactants is 7.5, while that of products n_p is 6, so that

$$\Delta n = n_p - n_r = -1.5$$

It may be taken as 2 cal for $2 \times 10^{-3} (\text{kcal}) \text{ deg}^{-1} \text{ mole}^{-1}$.

Hence,

$$\Delta H = \Delta E + R T \Delta n$$

$$-781 \text{ kcal} = \Delta E - [2 \times 10^{-3} (\text{kcal deg}^{-1} \text{ mole}^{-1}) \times 298 \times 1.5]$$

$$= \Delta E - 0.894$$

$$\text{or } \Delta E_{298} = (0.894 - 781) \text{ kcal}.$$

$$= -780.11 \text{ kcal.} \quad \text{This is the heat of reaction at constant volume.}$$

DIFFERENT TYPES OF ENTHALPY OF REACTION

The enthalpy changes accompanying chemical reactions are expressed in different ways, depending on the nature of the reaction. These are as follows:-

Heat (Enthalpy) of formation:- The change in enthalpy of a system when the moles of the substance is ~~completely~~ ^{formed from its elements} ~~burnt in excess of air or oxygen~~. If the reactants and products involved in the reaction are in standard state (298K and 1 atm), the enthalpy change is referred to as the standard heat of formation.

Example



Heat of formation is denoted by ΔH_f° .

Heat of Combustion, ΔH_c° :- The change in enthalpy of a system when one mole of the substance is completely burnt in excess of air or oxygen. It is referred to enthalpy of combustion.

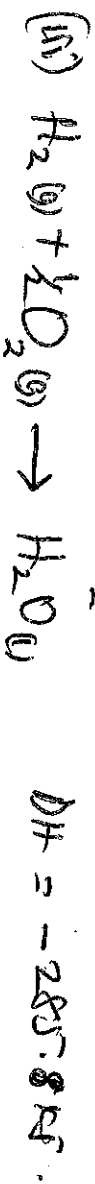
For example:- the enthalpy of combustion of methane is -210 kcal ($= 87.78 \text{ kJ}$). $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$ $\Delta H_c^\circ = -210 \text{ kcal}$. The enthalpy of combustion of a substance (ΔH_c°) is always -ve.

Question: Cal. the standard heat of formation of propane (C_3H_8) if its heat of combustion is -2220.2 kJ/mol . The heat of formation of $\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\text{l})$ are -393.5 and -285.8 kJ/mol respectively.

Solution:

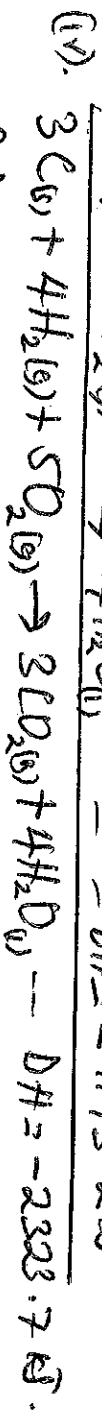
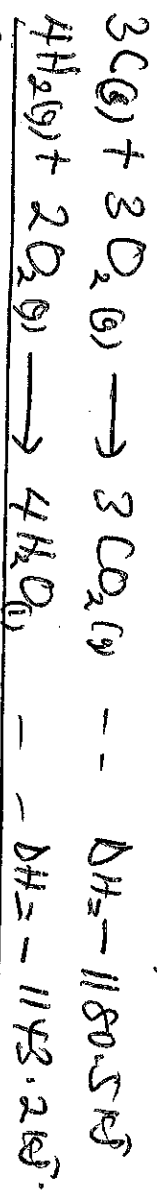
Solution:

like are given;

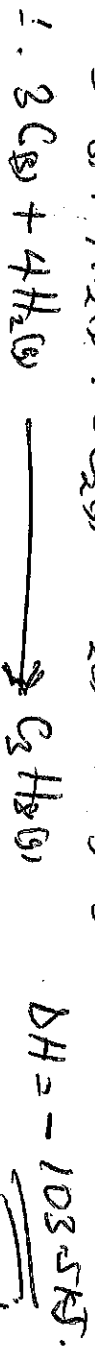


In order to solve the problem from the above equation we need to manipulate; the actual equation should be in the form $3\text{C}(s) + 4\text{H}_2(g) \rightarrow \text{C}_3\text{H}_8(g) \quad \Delta H = ?$

$\times (i)$ by 3 and (iii) by 4 and then adding up, we get;



Subtracting (i) from (iv) , to get,



$\therefore$ The enthalpy of formation of propane is $-103.5 \text{ kJ mol}^{-1}$.

Enthalpy of Solution:- The enthalpy of solution is defined as the change in enthalpy when one mole of a substance is dissolved in a specified quantity of solvent at a given temperature.

Enthalpy of Neutralisation:- is defined as the change in heat content (enthalpy) of the system when one ~~mol~~ gram equivalent of an acid is neutralised by one gram equivalent of a base and vice versa in dilute solution.



the heat of neutralisation of strong acid and strong base is -13.7 kcal , no matter whether acid or base is employed. Thus, heat of neutralisation of acid and base is merely the heat of formation of water from hydrogen and hydroxyl ions.

Heat (enthalpy) of sublimation: is the heat change for enthalpy change when one mole of a solid is directly converted into the gaseous state at a ~~constant~~ temperature below its melting point. ^{sublimation}

for example, the reaction of iodine, the enthalpy



Enthalpy of Transition: is the change in enthalpy which occurs when one mole of an element changes from one allotropic form to another.

for example the transition of diamond into amorphous carbon may be represented as,



HESS'S LAW OF CONSTANT HEAT SUMMATION:

It is known that ΔE and ΔH are functions of the state of the system. The heat evolved or absorbed in a given reaction must be independent of the manner in which the reaction is brought about. Thus it depends only on the initial and final states of the system and not the manner or the steps in which the change takes place. This generalization is known as Hess's law and may be stated as:-

If a chemical change can be made to take place in two or more different ways whether in one step or two or more steps, the amount of total heat change is same no matter by which method the change is brought about.

The law also follows as a mere consequence of the first law of thermodynamics.

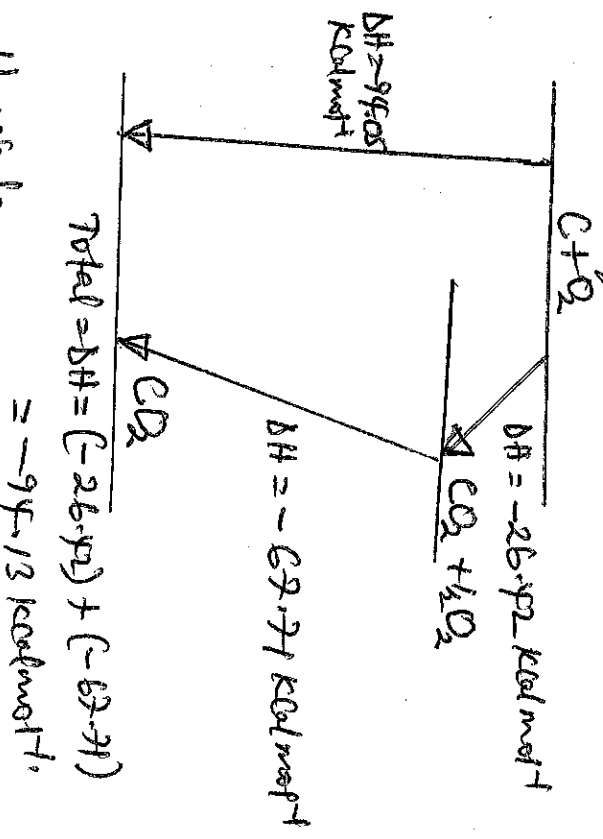
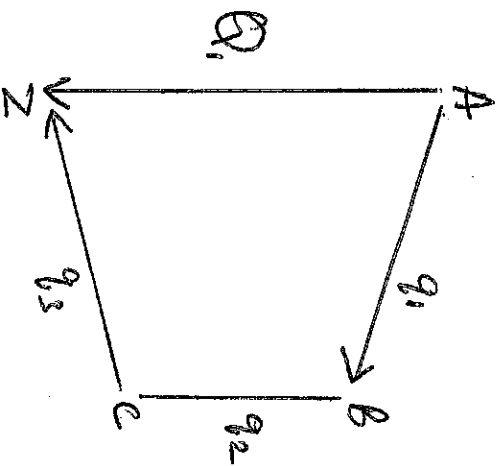
Let us suppose that a substance A can be changed to Z directly. (17)

where Q_1 is the heat evolved in the direct change. When the same change is brought about in stages:-



The total evolution of heat = $q_1 + q_2 + q_3 = Q_1$.

According to Hess's law $Q_1 = Q_2$. If it be so, let $Q_2 > Q_1$, then by transforming A to Z through stages and retransforming directly back to A, there would be gain of heat energy = $Q_2 - Q_1$. By repeating the process again and again an unlimited heat energy will be developed in an isolated system. This goes against the first law of thermodynamics. Hence Q_1 must be equal to Q_2 .



$\text{Total } \Delta H = (-26.42) + (-67.71)$
 $= -94.13 \text{ kcal/mol}^\circ$

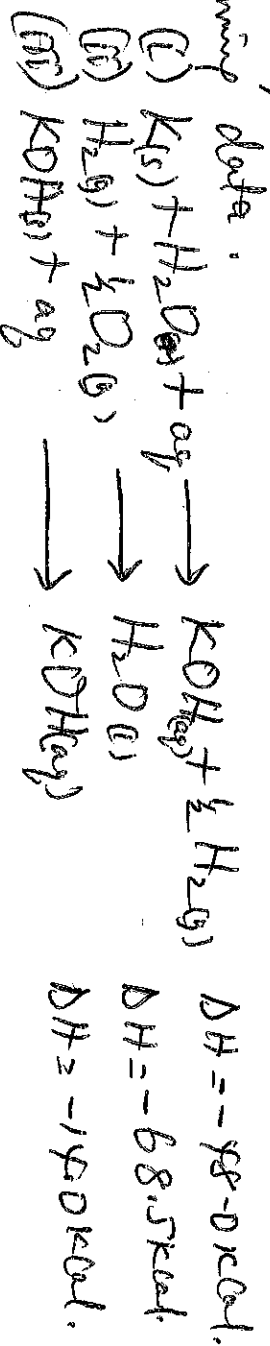
Illustration of Hess's law.

Applications of Hess's Law.

① Determination of heat of formation of substances which otherwise cannot be measured experimentally.

Substances like methane, benzene, carbon monoxide etc. cannot be prepared by uniting their elements, therefore it is not possible to measure the heat of formation of such compounds directly. These can be determined indirectly by using Hess's law.

Example: Calculate the heat of formation of KOH from the following data.



Solution:

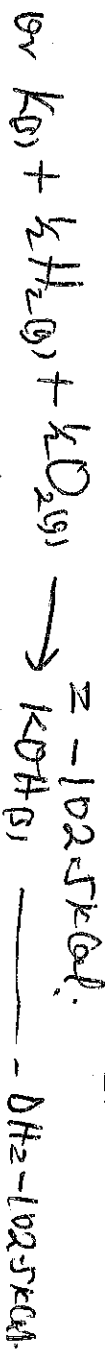


To find the target equation of formation above for ΔH , adding equation (i) and (ii) and subtracting (iii) will give:

$$\text{K(s)} + \text{H}_2\text{O(l)} + \text{H}_2\text{(g)} + \frac{1}{2}\text{O}_2\text{(g)} - \text{KOH(s)} \longrightarrow \text{KOH(aq)} + \frac{1}{2}\text{H}_2\text{(g)} + \text{H}_2\text{O(l)} - \text{KOH(s)}$$

$$\Delta H = -48.0 + (-68.5) - (-14.0)$$

$$= -102.5 \text{ kcal}$$



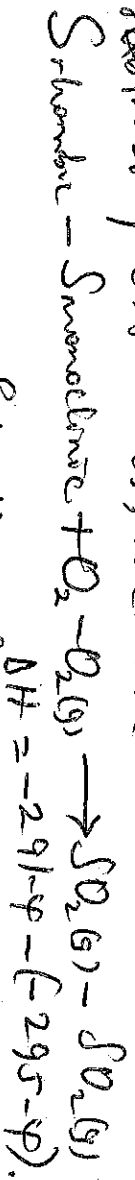
Thus, heat of formation of KOH is -102.5 kcal .

② Determination of Heat of Transition.

Example: The enthalpy of transition from monoclinic sulphur to rhombic sulphur can be calculated from the heat of combustion which are:-



Subtracting (ii) from (i), we have



$$\text{S(rhombic)} = \text{S(monoclinic)} \quad \Delta H = 4.0 \text{ kJ} //$$

Bond Energy:

When bond between two atoms formed, there is a release of energy. The same amount of energy is absorbed when the bond is broken.

The bond energy is defined as the average amount of energy required to break all bonds of a particular type in one mole of the substance. It is expressed in kJ mol^{-1} .

The bond energy of some common bonds are as follows:

Bond	Bond energy.
C-Cl	243 kJ mol^{-1}
C=O	$499.0 \text{ kJ mol}^{-1}$
C-H	$414.0 \text{ kJ mol}^{-1}$
O-H	$460.0 \text{ kJ mol}^{-1}$

Bond energy is the measure of strength of the bond. In other words, bond energy is the force with which the atoms are bonded together. It depends upon:-

- (i) Size of the atom
- (ii) Electronegativity
- (iii) Bond length.

When a bond is broken, the bond energy is +ve because heat is absorbed.